

ADITHYA BALASUBRAMANI

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Summary

AI & ML undergraduate with hands-on expertise in **Generative AI, Machine Learning, Data Engineering**, and **Cloud Computing** (AWS). Proficient in **Python programming** with experience building end-to-end ML pipelines, real-time AI-powered applications, and scalable cloud-based infrastructures. Demonstrated ability to develop and deploy production-grade AI solutions using LLMs, Transformer architectures, and modern data systems. Eager to contribute to innovative, real-world AI research and development.

Education

Vellore Institute of Technology (VIT), Chennai

2023 – Present

Bachelor of Technology in Computer Science and Engineering (AI & ML) — CGPA: 8.10

Chennai, India

The Velammal International School

2021 – 2023

Higher Secondary Education — Class XII: 85%

Chennai, India

Relevant Coursework

- Machine Learning
- Artificial Intelligence
- Data Structures
- Algorithms Analysis
- Database Management
- Cloud Computing
- Natural Language Processing
- Deep Learning

Technical Skills

Programming Languages: Python, Java, C, C++, SQL, HTML/CSS, JavaScript

AI/ML & Data Science: Scikit-learn, TensorFlow, PyTorch, LLMs, NLP, Generative AI, Transformers, Deep Learning

Data Engineering & Databases: PostgreSQL, SQL, NoSQL, FastAPI, Flask, Data Pipelines, ETL

Cloud Platforms & DevOps: AWS (Lambda, S3, CloudFront), Render, Docker, Git, GitHub Actions, Terraform, CI/CD

Frameworks & Tools: LangChain, LangGraph, React.js, Node.js, OpenCV, Tailwind CSS, VS Code

Projects

Personal AI Assistant | *Python, LangChain, LangGraph, FastAPI, AWS, React.js, PostgreSQL*

2025

- Developed a full-stack agentic **Generative AI** assistant using **LangChain/LangGraph** with stateful conversations and persistent **PostgreSQL**-backed memory for scalable data management.
- Built and deployed a **Python** FastAPI backend on **AWS Lambda** (Docker) with OAuth2/JWT authentication, achieving secure and low-latency AI interactions on **cloud infrastructure**.
- Implemented CI/CD pipelines with GitHub Actions and Terraform, deploying a React frontend via **AWS S3 + CloudFront** with automated, zero-downtime releases.

AI-Powered Fraud Detection System | *Python, Scikit-learn, FastAPI, SMOTE*

2024

- Developed a high-throughput fraud detection **data pipeline** in **Python** utilizing **Scikit-learn**, implementing SMOTE to address class imbalance across 100K+ transaction records.
- Engineered a RESTful microservice using **FastAPI** for real-time transaction monitoring, achieving under 200ms response latency for live **machine learning** inference.
- Optimized **ML model** performance to boost fraud detection rate by 30% over baseline while reducing false negatives by 22% through hyperparameter tuning.

Breast Cancer Detection using CNN | *Python, TensorFlow, Keras, OpenCV, Deep Learning*

2024

- Designed and trained a Convolutional Neural Network on 2,373 histopathological images using **TensorFlow/Keras** for binary classification of breast cancer.
- Achieved 89.6% test accuracy with AUC of 0.959, delivering 95.1% recall and 90.3% precision for malignant cases by prioritizing detection sensitivity.
- Applied advanced **data engineering** techniques including image augmentation, preprocessing pipelines, and dataset balancing to improve model robustness.

Certifications

Generative AI Using IBM Watsonx

2024

IBM Career Education Program

IBM

- Completed hands-on certification focused on **Generative AI** concepts, prompt engineering, and building AI-powered applications using IBM Watsonx cloud platform.